SIMULTANEOUS NULL CONTROL AND APPROXIMATE BOUNDARY TRACES FOR AN INTERSECTING HEAT CASCADE

SYLVAIN ERVEDOZA, PIERRE LISSY, YANNICK PRIVAT

Editorial note. Original author statement and full paired proof retained; source license and version reused. No mathematical modifications or added assumptions.

Editorial note. Disjoint boundary trace control cannot prescribe an exact terminal state. Public 2109 controls a nonlinear finite-dimensional shape derivative in a disjoint geometry; neither that target nor the prior obstruction is this mixed exact/approximate controllability theorem.

Allowed standard prerequisites (editorial).

- Classical heat equation and functional analysis. Weak and transposition solutions, Sobolev trace maps and parabolic regularity, Holmgren unique continuation, Hilbert-space range/kernel duality, standard heat observability and convex minimization. The displayed Hadamard derivative formula is a standard supplied input. Target-specific constructions remain in the original proof.

Let $T > 0$ denote a horizon of time, ω and Θ be two open subsets of $\mathbb{R}^d$, $d \in \mathbb{N}^*$, and $\xi \in L^2(0,T;L^2(\mathbb{R}^d))$.

For a bounded, connected and open set Ω of $\mathbb{R}^d$ of class $\mathscr{C}^2$, we consider the shape functional J_h, indexed by $h \in L^2(0,T;L^2(\omega))$, defined by

$$J_h(\Omega) = \frac{1}{2}\int_0^T \int_\Theta y_{\Omega,h}(t,x)^2\,dx\,dt, \tag{1}$$

where $y_{\Omega,h} \in L^2((0,T)\times\mathbb{R}^d)$ is defined on $(0,T)\times\Omega$ as the unique weak solution of

$$\left\{\begin{array}{rcll} \dfrac{\partial y}{\partial t} - \Delta y &=& \xi + h\mathbb{1}_\omega & \text{in } (0,T)\times\Omega,\\ y &=& 0 & \text{on } (0,T)\times\partial\Omega,\\ y\,(0,\cdot) &=& 0 & \text{in } \Omega, \end{array}\right. \tag{2}$$

($\mathbb{1}_\omega$ denotes the characteristic function of the set ω) extended by 0 outside $(0,T)\times\Omega$.

In the sequel, given $\boldsymbol{V} \in W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d)$, we introduce the family $\{\Omega_{\tau\boldsymbol{V}}\}_{\tau\in[0,\tau_{\boldsymbol{V}})}$ of domains defined by

$$\Omega_{\tau\boldsymbol{V}} = (\mathrm{Id} + \tau\boldsymbol{V})(\Omega_0).$$

Proposition 0.1 ([28, Proof of Prop. 1.1])**.** *Let $\xi \in L^2(0,T;L^2(\mathbb{R}^d))$ and $h \in L^2(0,T;L^2(\omega))$. For all $\boldsymbol{V} \in W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d)$, the mapping $\tau \mapsto J_h(\Omega_{\tau\boldsymbol{V}})$ is differentiable at $\tau = 0$ and*

$$\frac{d}{d\tau}\left(J_h\left(\Omega_{\tau\boldsymbol{V}}\right)\right)\Big|_{\tau=0} = \int_{\partial\Omega_0} \boldsymbol{V}\cdot\boldsymbol{n}\left(\int_0^T \partial_n y_0 \partial_n z_0\,dt\right)d\sigma, \tag{3}$$

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where the pair (y_0, z_0) solves the coupled forward-backward system

$$
(4)\qquad \begin{cases} \dfrac{\partial y_0}{\partial t} - \Delta y_0 &= \xi + h\mathbb{1}_\omega & \text{in } (0,T)\times\Omega_0,\\ y_0 &= 0 & \text{on } (0,T)\times\partial\Omega_0,\\ y_0(0,\cdot) &= 0 & \text{in } \Omega_0,\\ -\dfrac{\partial z_0}{\partial t} - \Delta z_0 &= \mathbb{1}_\Theta y_0 & \text{in } (0,T)\times\Omega_0,\\ z_0 &= 0 & \text{on } (0,T)\times\partial\Omega_0,\\ z_0(T,\cdot) &= 0 & \text{in } \Omega_0. \end{cases}
$$

($\mathbb{1}_\Theta$ denotes the characteristic function of the set Θ.)

Definition 0.2. Let $\xi \in L^2(0,T;L^2(\mathbb{R}^d))$. We say that the control function $h \in L^2(0,T;L^2(\omega))$ desensitizes J_h exactly at Ω_0 at the first order with respect to domain variations if

$$
(5)\qquad \forall\, \boldsymbol{V} \in W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d), \quad \frac{d}{d\tau}\left(J_h\left(\Omega_{\tau\boldsymbol{V}}\right)\right)\Big|_{\tau=0} = 0.
$$

Let $\mathcal{E}$ be a linear subspace of $W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d)$. We say that the control function $h \in L^2(0,T;L^2(\omega))$ exactly desensitizes J_h for $\mathcal{E}$ at Ω_0 at the first order with respect to domain variations if

$$
(6)\qquad \forall\, \boldsymbol{V} \in \mathcal{E}, \quad \frac{d}{d\tau}\left(J_h\left(\Omega_{\tau\boldsymbol{V}}\right)\right)\Big|_{\tau=0} = 0.
$$

Given $\varepsilon > 0$, we say that the control function $h \in L^2(0,T;L^2(\omega))$ ε-approximately desensitizes J_h at Ω_0 at the first order with respect to domain variations if

$$
(7)\qquad \forall\, \boldsymbol{V} \in W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d), \quad \left|\frac{d}{d\tau}\left(J_h\left(\Omega_{\tau\boldsymbol{V}}\right)\right)\Big|_{\tau=0}\right| \leq \varepsilon\|\boldsymbol{V}\|_{W^{3,\infty}(\mathbb{R}^d;\mathbb{R}^d)}.
$$

0.1. **Main results.** As we said, the aforementioned desensitizing problems will be strongly dependent on the relative geometry of the various sets ω, Θ and Ω_0, and in particular depending if the set $\omega\cap\Theta$ is empty or not, but we will always make the following minimal assumption on both sets ω and Θ:

$(\mathrm{H}_{\omega,\Theta})$ ω and Θ are two nonempty open subsets of Ω_0.

To be more precise, the first geometric setting we consider is the following:

$$
(8)\qquad \omega \text{ and } \Theta \text{ satisfy } (\mathrm{H}_{\omega,\Theta}), \quad \omega\cap\Theta = \emptyset, \quad \Theta \Subset \Omega_0, \text{ and } \Omega_0\setminus\overline{\Theta} \text{ is connected.}
$$

Proposition 0.3. *Assume the geometric setting* (8)*. Then, given any $(f_1,f_2) \in L^2(0,T;L^2(\partial\Omega_0))^2$, for any $\varepsilon > 0$ and $\xi \in L^2(0,T;L^2(\Omega_0))$, there exists a control function $h \in L^2(0,T;L^2(\omega))$ such that the solution (y,z) of*

$$
(9)\qquad \begin{cases} \dfrac{\partial y}{\partial t} - \Delta y &= \xi + h\mathbb{1}_\omega & \text{in } (0,T)\times\Omega_0,\\ y &= 0 & \text{on } (0,T)\times\partial\Omega_0,\\ y(0,\cdot) &= 0 & \text{in } \Omega_0,\\ -\dfrac{\partial z}{\partial t} - \Delta z &= \mathbb{1}_\Theta y & \text{in } (0,T)\times\Omega_0,\\ z &= 0 & \text{on } (0,T)\times\partial\Omega_0,\\ z(T,\cdot) &= 0 & \text{in } \Omega_0, \end{cases}
$$

satisfies

$$
(10)\qquad \|\partial_n y - f_1\|_{L^2(0,T;L^2(\partial\Omega_0))} + \|\partial_n z - f_2\|_{L^2(0,T;L^2(\partial\Omega_0))} \leq \varepsilon.
$$

Case $\omega \cap \Theta \neq \emptyset$. The second geometric setting we consider is the case

$$\omega \text{ and } \Theta \text{ satisfy } (\mathrm{H}_{\omega,\Theta}), \quad \omega \cap \Theta \neq \emptyset. \tag{11}$$

This case is more favorable since the control set ω meets the observation set Θ, and as we shall see afterward, not only do all previously established results remain true, but this also allows to prove the existence of even better desensitizing controls.

Proposition 0.4. *Assume the geometric setting* (11)*. Then, given any $(f_1, f_2) \in (L^2(0,T;L^2(\partial\Omega_0)))^2$, $y_T \in L^2(\Omega)$, any $\varepsilon > 0$ and $\xi \in L^2(0,T;L^2(\Omega_0))$, there exists a control function $h \in L^2(0,T;L^2(\omega))$ such that the solution (y,z) of* (9) *satisfies*

$$\|\partial_n y - f_1\|_{L^2(0,T;L^2(\partial\Omega_0))} + \|\partial_n z - f_2\|_{L^2(0,T;L^2(\partial\Omega_0))} + \|y(T) - y_T\|_{L^2(\Omega_0)} \leq \varepsilon. \tag{12}$$

Besides, if the source term $\xi \in L^2(0,T;L^2(\Omega_0))$ is null-controllable in the sense that there exists $h_{\mathrm{nc}} \in L^2(0,T;L^2(\omega))$ such that the solution y_{nc} of

$$\begin{cases} \dfrac{\partial y_{\mathrm{nc}}}{\partial t} - \Delta y_{\mathrm{nc}} &= \xi + h_{\mathrm{nc}} \mathbb{1}_\omega & \text{in } (0,T)\times\Omega_0, \\ y_{\mathrm{nc}} &= 0 & \text{on } (0,T)\times\partial\Omega_0, \\ y_{\mathrm{nc}}(0,\cdot) &= 0 & \text{in } \Omega_0, \end{cases} \tag{13}$$

satisfies

$$y_{\mathrm{nc}}(T) = 0 \text{ in } \Omega_0, \tag{14}$$

then, given any $(f_1, f_2) \in L^2(0,T;L^2(\partial\Omega_0))^2$, for any $\varepsilon > 0$, there exists a control function $h \in L^2(0,T;L^2(\omega))$ such that the solution (y,z) of (9) *satisfies*

$$\|\partial_n y - f_1\|_{L^2(0,T;L^2(\partial\Omega_0))} + \|\partial_n z - f_2\|_{L^2(0,T;L^2(\partial\Omega_0))} \leq \varepsilon. \tag{15}$$

and

$$y(T) = 0 \text{ in } \Omega_0. \tag{16}$$

Remark 0.5. Determining a control function h such that the solution y_{nc} of (13) satisfies (14) is the well-known null-controllability problem with source term for the heat equation. This issue has been much investigated. By using duality arguments, this issue can be recast in terms of a so-called "observability inequality". More precisely, one can deduce from [17, Lem. 2.1] that there exists $C > 0$ such that

$$\left\| \exp\Big(-\frac{C}{T-t}\Big)\varphi(t) \right\|_{L^2(0,T;L^2(\Omega_0))} \leq C \|\varphi\|_{L^2(0,T;L^2(\omega))}, \tag{17}$$

where φ denotes the solution of the backward adjoint system

$$\begin{cases} -\dfrac{\partial \varphi}{\partial t} - \Delta \varphi &= 0 & \text{in } (0,T)\times\Omega_0, \\ \varphi &= 0 & \text{on } (0,T)\times\partial\Omega_0, \\ \varphi(T,\cdot) &= \varphi^T & \text{in } \Omega_0. \end{cases}$$

From the observability inequality (17), we can deduce that for any $\xi \in L^2(0,T;L^2(\Omega_0))$ with $e^{C/(T-t)}\xi \in L^2(0,T;L^2(\Omega_0))$, one can find a null control h_{nc} to (13), i.e.a function $h_{\mathrm{nc}} \in L^2(0,T;L^2(\omega))$ such that the solution y_{nc} of (13) satisfies (14), as explained in [17, Proof of Th. 2.1].

0.2. **Proof of Proposition 0.3.** Proposition 0.3 can be recast in an abstract form into the problem: show that

$$\overline{\mathrm{Ran}\,\mathcal{L}} = (L^2(0,T;L^2(\partial\Omega_0)))^2,$$

where $\mathcal{L}$ is the operator defined for $h \in L^2(0,T;L^2(\omega))$ by

$$\mathcal{L}h = (\partial_n y_h, \partial_n z_h) \in L^2((0,T)\times\partial\Omega_0)^2,$$

where (y_h, z_h) solves

$$\left\{\begin{array}{rcll} \dfrac{\partial y_h}{\partial t} - \Delta y_h &=& h\mathbb{1}_\omega & \text{in } (0,T)\times\Omega_0,\\ y_h &=& 0 & \text{on } (0,T)\times\partial\Omega_0,\\ y_h(0,\cdot) &=& 0 & \text{in } \Omega_0,\\ -\dfrac{\partial z_h}{\partial t} - \Delta z_h &=& \mathbb{1}_\Theta y_h & \text{in } (0,T)\times\Omega_0,\\ z_h &=& 0 & \text{on } (0,T)\times\partial\Omega_0,\\ z_h(T,\cdot) &=& 0 & \text{in } \Omega_0. \end{array}\right. \tag{18}$$

Notice that $\mathcal{L}$ is bounded, thanks to [15, Th. 5, p. 382] and the continuity of the operator $f \in H^2(\Omega_0) \mapsto \partial_n f \in L^2((0,T)\times\partial\Omega_0)$. Therefore, by standard arguments from functional analysis, Proposition 0.3 is equivalent to showing the dual property $\operatorname{Ker}\mathcal{L}^* = \{0\}$ (we refer for instance to [7, Cor. 2.18]). Using the arguments developed in [28, Prop. 2.1], one can compute explicitly $\mathcal{L}^*$ and deduce that $\operatorname{Ker}\mathcal{L}^* = \{0\}$ is equivalent to the following unique continuation problem: If $(g_1, g_2) \in (L^2(0,T;L^2(\partial\Omega_0)))^2$, and (ψ, φ) solves

$$\left\{\begin{array}{rcll} -\dfrac{\partial \psi}{\partial t} - \Delta \psi &=& \mathbb{1}_\Theta \varphi & \text{in } (0,T)\times\Omega_0,\\ \psi &=& g_1 & \text{on } (0,T)\times\partial\Omega_0,\\ \psi(T,\cdot) &=& 0 & \text{in } \Omega_0,\\ \dfrac{\partial \varphi}{\partial t} - \Delta \varphi &=& 0 & \text{in } (0,T)\times\Omega_0,\\ \varphi &=& g_2 & \text{on } (0,T)\times\partial\Omega_0,\\ \varphi(0,\cdot) &=& 0 & \text{in } \Omega_0, \end{array}\right. \tag{19}$$

then, we have the following unique continuation property:

$$\psi = 0 \text{ in } (0,T)\times\omega \implies g_1 = g_2 = 0. \tag{20}$$

We now prove this unique continuation property. Let $(g_1, g_2) \in (L^2(0,T;L^2(\partial\Omega_0)))^2$ be such that the solution (ψ, φ) of (19) satisfies $\psi = 0$ in $(0,T)\times\omega$. From [27, (15.17–18), p. 86], we notably infer that $(\psi, \varphi) \in L^2((0,T)\times\Omega_0)$.

We first work in the set $(0,T)\times(\Omega_0 \setminus \overline{\Theta})$. There, ψ satisfies the usual backward heat equation

$$-\frac{\partial \psi}{\partial t} - \Delta\psi = 0 \quad \text{in } (0,T)\times(\Omega_0\setminus\overline{\Theta}),$$

where we do not specify any initial or boundary conditions. Therefore, since $\omega \subset \Omega_0 \setminus \overline{\Theta}$ and $\Omega_0 \setminus \overline{\Theta}$ is connected thanks to Assumption (8), using the Holmgren theorem for the heat equation (see e.g.[21, §8.6 & Th. 8.6.5]), we thus infer that $\psi = 0$ in $(0,T)\times(\Omega_0 \setminus \overline{\Theta})$. In particular, using [34, Prop. 7.1.3], since ψ the unique weak solution of (19) in the sense of [34, Def. 7.1.2, p. 342], for any $\zeta \in H^2(\Omega_0)\cap H_0^1(\Omega_0)$ such that $\Delta\zeta \in H_0^1(\Omega_0)$, for any $t \in [0,T]$, we have

$$\int_0^t \int_{(0,T)\times\partial\Omega_0} g_1 \partial_n \zeta \, d\sigma dt = 0.$$

Differentiating this inequality with respect to t (which is possible because

$$g_1(\cdot,x)\partial_n\zeta(x) \in L^2(0,T) \quad \text{for almost all } x \in (0,T)\times\partial\Omega_0,$$

since the Neumann trace operator $f \in H^2(\Omega_0) \mapsto \partial_n f \in L^2(\partial\Omega_0)$ is well-defined), we obtain that for any $t \in [0,T]$, we have

$$\int_{\partial\Omega_0} g_1(t,\cdot)\partial_n\zeta \, d\sigma = 0.$$

Since $\{\zeta \in H^2(\Omega_0) \cap H_0^1(\Omega_0) \mid \Delta\zeta \in H_0^1(\Omega_0)\}$ is dense in $H^2(\Omega_0) \cap H_0^1(\Omega_0)$ for the H^2-norm, using the continuity of the Neumann trace operator

$$f \in H^2(\Omega_0) \longmapsto \partial_n f \in L^2((0,T) \times \partial\Omega_0),$$

we deduce that for any $\zeta \in H^2(\Omega_0) \cap H_0^1(\Omega_0)$, we have, for any $t \in [0,T]$,

$$\int_{\partial\Omega_0} g_1(t,\cdot)\partial_n\zeta d\sigma = 0.$$

Since the extended trace operator

$$f \in H^2(\Omega) \longmapsto (f_{|\partial\Omega_0}, \partial_n f_{|\partial\Omega_0}) \in H^{3/2}(\partial\Omega_0) \times H^{1/2}(\partial\Omega_0)$$

is surjective, taking into account that $0 \in H^{3/2}(\partial\Omega_0)$, we deduce that for any $h \in H^{1/2}(\partial\Omega_0)$, we have, for any $t \in [0,T]$,

$$\int_{\partial\Omega_0} g_1(t,\cdot)h d\sigma = 0.$$

Since $H^{1/2}(\partial\Omega_0)$ is dense in $L^2(\partial\Omega_0)$ for the $L^2(\partial\Omega_0)$-norm, we deduce that $g_1 = 0$ on $(0,T) \times \partial\Omega_0$, so that $\psi \equiv 0$ on $\Omega_0 \setminus \overline{\Theta}$ in the classical sense. Notably, $\partial_n\psi = 0$ on the whole boundary $\partial\Omega_0$ since $\Theta \Subset \Omega_0$.

We next remark that, by standard local regularity results for solutions of the heat equation, the function φ is smooth away from the boundary $\{0,T\} \times \partial\Omega_0$ (of class $\mathscr{C}^\infty((0,T) \times \Omega_0)$). Accordingly, $\mathbb{1}_\Theta\varphi$ belongs to $L^2(0,T;L^2(\Omega))$, and by standard maximal regularity results, the solution ψ belongs to

$$L^2(0,T;H^2_{\mathrm{loc}}(\Omega_0)) \cap H^1(0,T;L^2_{\mathrm{loc}}(\Omega_0)).$$

In view of this regularity, and due to the fact that $\psi = 0$ in a neighbourhood of $(0,T) \times \partial\Omega_0$, we can multiply equation $(19)_{(1)}$ by φ and perform several integration by parts, using $g_1 = \partial_n\psi = 0$ on $\partial\Omega_0$, and $\psi = 0$ in $(0,T) \times (\Omega_0 \setminus \overline{\Theta})$:

$$\begin{aligned}
\int_0^T \int_\Theta |\varphi|^2 \, dx \, dt &= \int_0^T \int_{\Omega_0} \varphi\Big(-\frac{\partial\psi}{\partial t} - \Delta\psi\Big) \, dx \, dt \\
&= -\int_{\Omega_0} \varphi(\cdot,x)\psi(\cdot,x) \, dx\Big|_0^T - \int_0^T \int_{\partial\Omega_0} \varphi\partial_n\psi d\sigma dt \\
&\qquad + \int_0^T \int_{\partial\Omega_0} \partial_n\varphi\psi d\sigma dt + \int_0^T \int_{\Omega_0} \Big(\frac{\partial\varphi}{\partial t} - \Delta\varphi\Big)\psi \, dx \, dt \\
&= 0.
\end{aligned}$$

Therefore, $\varphi = 0$ in $(0,T) \times \Theta$, and by the classical unique continuation properties for the heat equation, $\varphi = 0$ in $(0,T) \times \Omega_0$, and in particular $g_2 = 0$. This concludes the proof of (20) for the solutions of (19), which proves that $\operatorname{Ker}\mathcal{L}^* = \{0\}$. Hence, Proposition 0.3 follows.

0.3. **Proof of Proposition 0.4.** Similarly to the proof of Proposition 0.3, we reformulate the first part of Proposition 0.4, i.e.the approximate controllability property (12), as the density of the range of the operator $\mathcal{L}$ defined for $h \in L^2(0,T;L^2(\omega))$ with values in $(L^2(0,T;L^2(\partial\Omega_0)))^2 \times L^2(\Omega_0)$ by

$$\mathcal{L}(h) = (\partial_n y, \partial_n z, y(T)),$$

where (y,z) is the solution of (18). Arguments similar to the one developed in the proof of Proposition 0.3 imply that $\mathcal{L}$ is a bounded operator. By using similar classical arguments resting upon duality, as in the proof of Proposition 0.3, one has $\overline{\operatorname{Ran}\mathcal{L}} = (L^2(0,T;L^2(\partial\Omega_0)))^2 \times L^2(\Omega_0)$ if and only if $\operatorname{Ker}\mathcal{L}^* = \{0\}$ (we refer for instance to [7, Cor. 2.18]). Using the arguments developed in [28, Prop. 2.1] together with [12, Th. 2.43, p. 56] (in order to take into account the term $y(T)$

in the definition of $\mathcal{L}$), one can compute explicitly $\mathcal{L}^*$ and deduce that $\operatorname{Ker}\mathcal{L}^* = \{0\}$ is equivalent to the following unique continuation problem: if $(g_1, g_2, \psi_T) \in (L^2(0,T;L^2(\partial\Omega_0)))^2 \times L^2(\Omega_0)$, and (ψ, φ) solves

$$\left\{\begin{array}{rcll} -\dfrac{\partial\psi}{\partial t} - \Delta\psi &=& \mathbb{1}_\Theta\varphi & \text{in } (0,T)\times\Omega_0,\\ \psi &=& g_1 & \text{on } (0,T)\times\partial\Omega_0,\\ \psi(T,\cdot) &=& \psi_T & \text{in } \Omega_0,\\ \dfrac{\partial\varphi}{\partial t} - \Delta\varphi &=& 0 & \text{in } (0,T)\times\Omega_0,\\ \varphi &=& g_2 & \text{on } (0,T)\times\partial\Omega_0,\\ \varphi(0,\cdot) &=& 0 & \text{in } \Omega_0, \end{array}\right. \tag{21}$$

then, we have the following unique continuation property:

$$\psi = 0 \text{ in } (0,T)\times\omega \implies g_1 = g_2 = 0 \text{ and } \psi_T = 0. \tag{22}$$

Let us then take $(g_1, g_2, \psi_T) \in (L^2(0,T;L^2(\partial\Omega_0)))^2 \times L^2(\Omega_0)$, (ψ,φ) solving (21) with $\psi = 0$ in $(0,T)\times\omega$. Then the equation (21) on ψ implies that $\varphi = 0$ in $(0,T)\times(\omega\cap\Theta)$. The classical unique continuation property for the heat equation (for instance, by the Holmgren uniqueness theorem, see e.g.[21, §8.6 & Th. 8.6.5]) then applies to φ and $\varphi = 0$ in $(0,T)\times\Omega_0$. Following arguments similar to the proof of Proposition Proposition 0.3, we deduce that $g_2 = 0$, and that ψ solves the heat equation

$$-\frac{\partial\psi}{\partial t} - \Delta\psi = 0, \quad \text{in } (0,T)\times\Omega_0.$$

Since $\psi = 0$ in $(0,T)\times\omega$, we immediately deduce again, from the Holmgren uniqueness theorem (see e.g.[21, §8.6 & Th. 8.6.5]), that $\psi = 0$ in $(0,T)\times\Omega_0$. Since $\psi \in C^0([0,T], H^{-1}(\Omega_0))$ (see e.g.[34, Prop. 7.1.3]), the initial datum ψ_T vanishes as well. Following arguments similar to the proof of Proposition Proposition 0.3, we deduce that $g_1 = 0$.

This finishes the proof of the unique continuation property (22) for solutions of (21), hence the proof of the first part of Proposition 0.4, i.e.of the approximate controllability property (12).

Let us now focus on the proof of the null-controllability property when $\xi \in L^2(0,T;L^2(\Omega_0))$ can be steered to 0 using a control $h_{\rm nc} \in L^2(0,T;L^2(\omega))$, in the sense that the solution $y_{\rm nc}$ of (13) satisfies (14).

Then, for any $\varepsilon > 0$ and f_1, f_2 in $L^2(0,T;L^2(\partial\Omega_0))$, we look for y, z and h respectively as $y = y_{\rm nc} + y_1$, $z = z_1$ and $h = h_{\rm nc} + h_1$, where (y_1, z_1) satisfies

$$\left\{\begin{array}{rcll} \dfrac{\partial y_1}{\partial t} - \Delta y_1 &=& h_1\mathbb{1}_\omega & \text{in } (0,T)\times\Omega_0,\\ y_1 &=& 0 & \text{on } (0,T)\times\partial\Omega_0,\\ y_1(0,\cdot) &=& 0 & \text{in } \Omega_0,\\ -\dfrac{\partial z_1}{\partial t} - \Delta z_1 &=& \mathbb{1}_\Theta(y_{\rm nc} + y_1) & \text{in } (0,T)\times\Omega_0,\\ z_1 &=& 0 & \text{on } (0,T)\times\partial\Omega_0,\\ z_1(T,\cdot) &=& 0 & \text{in } \Omega_0, \end{array}\right. \tag{23}$$

and

$$\begin{cases} \|\partial_n y_1 - \tilde{f}_1\|_{L^2(0,T;L^2(\partial\Omega_0))} + \|\partial_n z_1 - f_2\|_{L^2(0,T;L^2(\partial\Omega_0))} \leq \varepsilon,\\ y_1(T) = 0 \text{ in } \Omega_0, \end{cases} \tag{24}$$

where

$$\tilde{f}_1 = f_1 - \partial_n y_{\rm nc}. \tag{25}$$

In order to do that, we consider the functional K_ε defined for

$$(g_1, g_2, \psi_T) \in (L^2(0,T;L^2(\partial\Omega_0)))^2 \times L^2(\Omega_0)$$

by

$$K_\varepsilon(g_1,g_2,\psi_T) = \frac{1}{2}\int_0^T \int_\omega |\psi(t,x)|^2\,dx\,dt + \int_0^T \int_{\partial\Omega_0} \left(\tilde{f}_1 g_1 + f_2 g_2\right) d\sigma dt + \int_0^T \int_{\Omega_0} \mathbb{1}_\Theta y_0 \varphi + \varepsilon \|(g_1,g_2)\|_{(L^2(0,T;L^2(\partial\Omega_0)))^2},$$

where (ψ,φ) is the solution of (21) corresponding to (g_1,g_2,ψ_T).

Then, we endow the set

$$X_0 = (L^2(0,T;L^2(\partial\Omega_0)))^2 \times L^2(\Omega_0)$$

with the norm

$$\|(g_1,g_2,\psi_T)\|_{\rm obs}^2 = \int_0^T \int_\omega |\psi(t,x)|^2\,dx\,dt + \int_0^T \int_{\partial\Omega_0} \left(|g_1|^2 + |g_2|^2\right) d\sigma dt,$$

where ψ is the solution of (21). The fact that this defines a norm comes from the unique continuation property (22). Then, we define

$$X_{\rm obs} = \overline{X_0}^{\|\cdot\|_{\rm obs}}.$$

We claim that K_ε can be extended continuously to $X_{\rm obs}$. Indeed, let us emphasize that for $(g_1,g_2,\psi_T) \in X_{\rm obs}$, $\psi|_{(0,T)\times\omega}$ is well defined by density as an element of $L^2(0,T;L^2(\omega))$ since the function $(g_1,g_2,\psi_T) \in X_0 \mapsto \psi|_{(0,T)\times\omega}$ is well-defined on X_0 and continuous (by construction) for the topology of $X_{\rm obs}$, and that we have the following straightforward estimate of the solution φ of (21): there exists $C>0$ such that for any $(g_1,g_2,\psi_T) \in X_0$,

$$\|\varphi\|_{L^2(0,T;L^2(\Omega_0))} \le C\|g_2\|_{L^2(0,T;L^2(\partial\Omega_0))} \le C\|(g_1,g_2,\psi_T)\|_{\rm obs}. \tag{26}$$

Now, given $(g_1,g_2,\psi_T) \in X_0$, ψ satisfies the backward heat equation (21). We then introduce the solution ψ_1 of

$$\left\{\begin{array}{rcll} -\dfrac{\partial\psi_1}{\partial t} - \Delta\psi_1 &=& 0 & \text{in } (0,T)\times\Omega_0,\\ \psi_1 &=& g_1 & \text{on } (0,T)\times\partial\Omega_0,\\ \psi_1\,(T,\cdot) &=& 0 & \text{in } \Omega_0, \end{array}\right.$$

which belongs to $L^2(0,T;L^2(\Omega_0))$, with

$$\|\psi_1\|_{L^2(0,T;L^2(\Omega_0))} \le C\|g_1\|_{L^2(0,T;L^2(\partial\Omega_0))}.$$

Next, $\psi-\psi_1$ satisfies the heat equation with homogeneous Dirichlet boundary conditions, so that classical Carleman estimates, see for instance [34, Th. 9.4.1] (after having bounded the weight function $\alpha(x)$ by some constant $C>0$ from above and by 1 from below) implies

$$\|e^{-C/(t(T-t))}(\psi-\psi_1)\|_{L^2(0,T;L^2(\Omega_0))} \le C\|\psi-\psi_1\|_{L^2(0,T;L^2(\omega))} + C\|-\partial_t\psi - \Delta\psi\|_{L^2(0,T;L^2(\Omega_0))}.$$

Using that the backward heat equation is well posed, standard energy estimates show that

$$\|e^{-C/(T-t)}(\psi-\psi_1)\|_{L^2(0,T;L^2(\Omega_0))} \le C\|\psi-\psi_1\|_{L^2(0,T;L^2(\omega))} + C\|-\partial_t\psi - \Delta\psi\|_{L^2(0,T;L^2(\Omega_0))}.$$

In particular,

$$\begin{aligned}\|e^{-C/(T-t)}\psi\|_{L^2(0,T;L^2(\Omega_0))} &\leq C\|\psi\|_{L^2(0,T;L^2(\omega))} + C\|\varphi\|_{L^2(0,T;L^2(\Omega_0))} \\ &\quad + C\|g_1\|_{L^2(0,T;L^2(\partial\Omega_0))} \\ &\leq C\|(g_1,g_2,\psi_T)\|_{\rm obs}.\end{aligned}$$

Hence, by density of X_0 in $X_{\rm obs}$, to each $(g_1,g_2,\psi_T)\in X_{\rm obs}$, we can associate a solution (ψ,φ) to

$$\left\{\begin{array}{rcll} -\dfrac{\partial\psi}{\partial t}-\Delta\psi &=& \mathbb{1}_\Theta\varphi & \text{in } (0,T)\times\Omega_0,\\ \psi &=& g_1 & \text{on } (0,T)\times\partial\Omega_0,\\ \dfrac{\partial\varphi}{\partial t}-\Delta\varphi &=& 0 & \text{in } (0,T)\times\Omega_0,\\ \varphi &=& g_2 & \text{on } (0,T)\times\partial\Omega_0,\\ \varphi(0,\cdot) &=& 0 & \text{in } \Omega_0,\end{array}\right.$$

such that for all $T'<T$, $\psi\in L^2(0,T';L^2(\Omega_0))$. We can thus apply the unique continuation property (22) with T replaced by T', using the same strategy as before relying on Holmgren's uniqueness theorem, and we obtain that if, for $(g_1,g_2,\psi_T)\in X_{\rm obs}$ we have $\psi|_{(0,T')\times\omega}=0$ then φ and ψ vanishes identically on $(0,T')$. Since $T'\in(0,T)$ is arbitrary, we deduce that the unique continuation property (22) extends to $(g_1,g_2,\psi_T)\in X_{\rm obs}$.

Classical contradiction arguments relying on (22) (see e.g.[14]) then give that

$$\liminf_{\|(g_1,g_2,\psi_T)\|_{\rm obs}\to\infty}\frac{K_\varepsilon(g_1,g_2,\psi_T)}{\|(g_1,g_2,\psi_T)\|_{\rm obs}}\geq\varepsilon. \tag{27}$$

Indeed, let us prove (27) by contradiction. Assume that there exists a sequence $(g_{1,n},g_{2,n},\psi_{T,n})\in X_{\rm obs}$ such that

$$\begin{cases}\rho_n:=\|(g_{1,n},g_{2,n},\psi_{T,n})\|_{\rm obs}\to\infty \text{ as } n\to\infty,\\ \alpha:=\limsup_{n\to\infty}\dfrac{K_\varepsilon(g_{1,n},g_{2,n},\psi_{T,n})}{\|(g_{1,n},g_{2,n},\psi_{T,n})\|_{\rm obs}}<\varepsilon.\end{cases} \tag{28}$$

We first renormalize the corresponding functions, and set

$$(\tilde g_{1,n},\tilde g_{2,n},\tilde\psi_{T,n})=\frac{(g_{1,n},g_{2,n},\psi_{T,n})}{\|(g_{1,n},g_{2,n},\psi_{T,n})\|_{\rm obs}},$$

so that

$$\|(\tilde g_{1,n},\tilde g_{2,n},\tilde\psi_{T,n})\|_{\rm obs}=1. \tag{29}$$

Therefore, there exists $(\tilde g_1,\tilde g_2,\tilde\psi_T)$ in $X_{\rm obs}$ such that

$$(\tilde g_{1,n},\tilde g_{2,n},\tilde\psi_{T,n}) \text{ weakly converges to } (\tilde g_1,\tilde g_2,\tilde\psi_T) \text{ in } X_{\rm obs} \text{ as } n\to\infty. \tag{30}$$

According to (28), we have

$$\begin{aligned}\frac12(\rho_n)^2\int_0^T\int_\omega|\tilde\psi_n|^2\,dx\,dt+\rho_n\bigg(\int_0^T\int_{\partial\Omega_0}\big(\tilde f_1\tilde g_{1,n}+f_2\tilde g_{2,n}\big)\,d\sigma dt\\ +\int_0^T\int_{\Omega_0}\chi_\Theta y_0\tilde\varphi_n+\varepsilon\|(\tilde g_{1,n},\tilde g_{2,n})\|_{(L^2(0,T;L^2(\partial\Omega_0)))^2}\bigg)\leq\alpha\rho_n.\end{aligned} \tag{31}$$

By (26) and (30), the sequence

$$\bigg(\int_0^T\int_{\partial\Omega_0}\big(\tilde f_1\tilde g_{1,n}+f_2\tilde g_{2,n}\big)\,d\sigma dt+\int_0^T\int_{\Omega_0}\chi_\Theta y_0\tilde\varphi_n \\ +\varepsilon\|(\tilde g_{1,n},\tilde g_{2,n})\|_{(L^2(0,T;L^2(\partial\Omega_0)))^2}\bigg)_{n\in\mathbb{N}}$$

is uniformly bounded in n. Therefore, dividing by ρ_n and using that $\rho_n \to +\infty$ when $n \to \infty$, we infer

$$\lim_{n\to\infty}\left(\int_0^T\int_\omega |\tilde{\psi}_n|^2\,dx\,dt\right) = 0,$$

hence

$$\tilde{\psi} = 0 \text{ in } (0,T)\times\omega.$$

We also deduce, according to (29), that

$$\lim_{n\to\infty} \|(\tilde{g}_{1,n},\tilde{g}_{2,n})\|_{(L^2(0,T;L^2(\partial\Omega_0)))^2} = 1.$$

Furthermore, using the unique continuation property (22), which has been shown to be also valid for elements of X_{obs}, we deduce that

$$(\tilde{g}_1,\tilde{g}_2,\tilde{\psi}_T) = (0,0,0).$$

In particular, according to (30) and the equation satisfied by $\tilde{\varphi}$,

$$\lim_{n\to\infty}\left(\int_0^T\int_{\partial\Omega_0}\left(\tilde{f}_1\tilde{g}_{1,n}+f_2\tilde{g}_{2,n}\right)d\sigma dt + \int_0^T\int_{\Omega_0}\chi_\Theta y_0\tilde{\varphi}_n \right.$$
$$\left. + \varepsilon\|(\tilde{g}_{1,n},\tilde{g}_{2,n})\|_{(L^2(0,T;L^2(\partial\Omega_0)))^2}\right) = \varepsilon,$$

which is in contradiction with (31), since α was assumed to be smaller than ε by (28). This concludes the proof of the coercivity estimate (27).

Therefore, since the functional K_ε is also obviously strictly convex on X_{obs}, it admits a unique minimizer $(g_1^*,g_2^*,\psi_T^*) \in X_{\text{obs}}$. Writing the Euler-Lagrange equation satisfied by the minimizer gives that, setting

$$h_1 = \psi^* \quad \text{in } (0,T)\times\omega,$$

the corresponding solution (y_1,z_1) of (23) satisfies (24). We refer for instance to [14, Th. 1.2 and its proof] for further details about the approach considered here. To conclude the proof of Proposition 0.4, we set $(y,z) = (y_1+y_{\text{nc}},z_1)$. Since (y_1,z_1) satisfies (23) and y_{nc} satisfies (13), (y,z) is a solution of (9), verifying moreover (15) and (16), thanks to (24), (25) and (14).

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